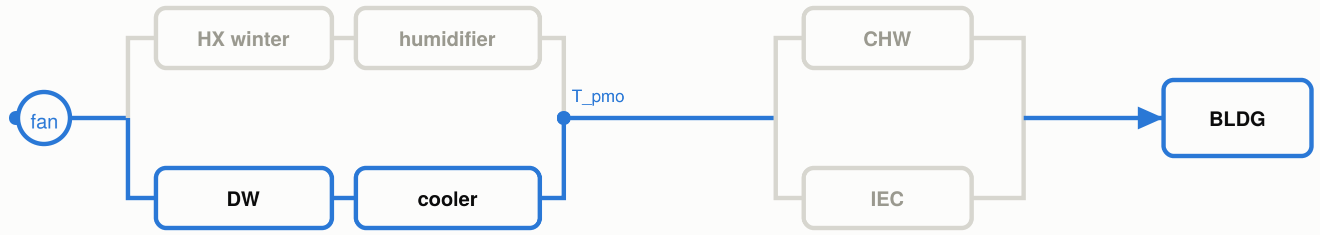
Topology variants — one per load quadrant and intensity tier

quadrants named by the processes they require; the tier says which components can deliver them

M1 cooling + dehumidification

tier: mild · geothermal water: ≥ 66 °C

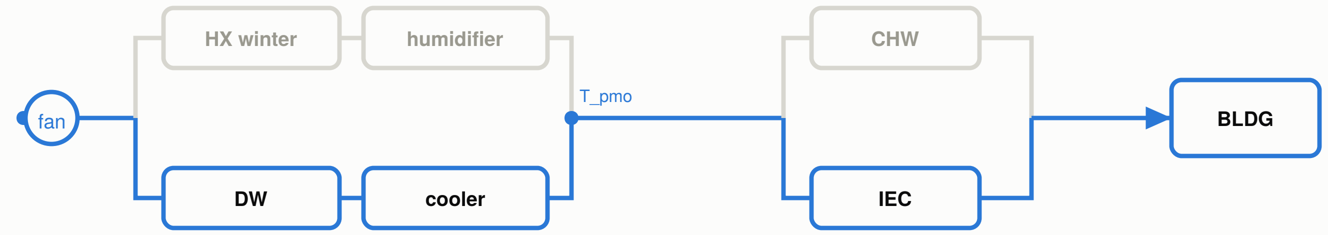


HX / hot water · scavenging loop

wheel dries; recuperative cooler trims the sensible load

M2 cooling + dehumidification

tier: moderate · geothermal water: ≥ 66 °C

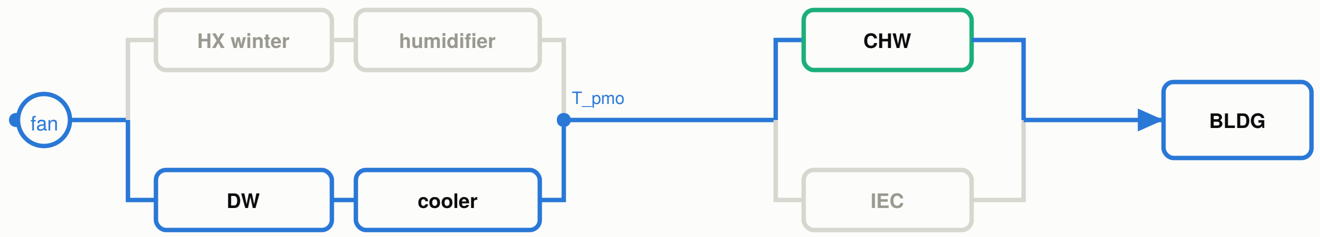


HX / hot water · scavenging loop

drying first collapses the wet-bulb, so the IEC bites deep

M3 cooling + dehumidification

tier: harsh · geothermal water: ≥ 79 °C

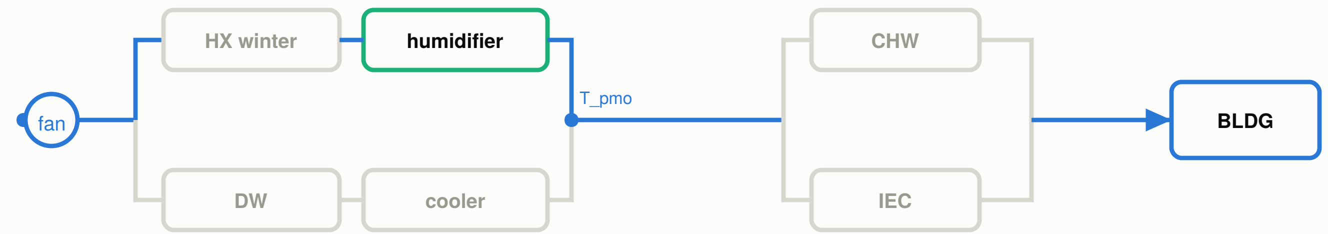


HX / hot water · ABS + WCT · chilled water · scavenging loop

chilled water for the sensible peak; wheel still carries the latent

M4 cooling + humidification

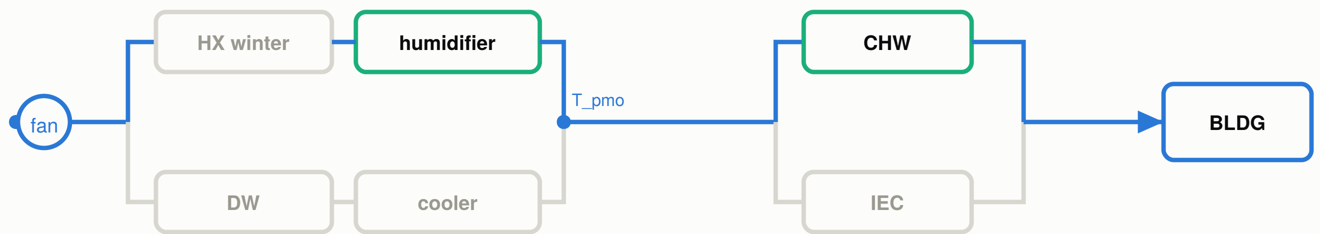
tier: mild · geothermal water: none



adiabatic humidification IS direct evaporative cooling

M5 cooling + humidification

tier: harsh · geothermal water: ≥ 79 °C

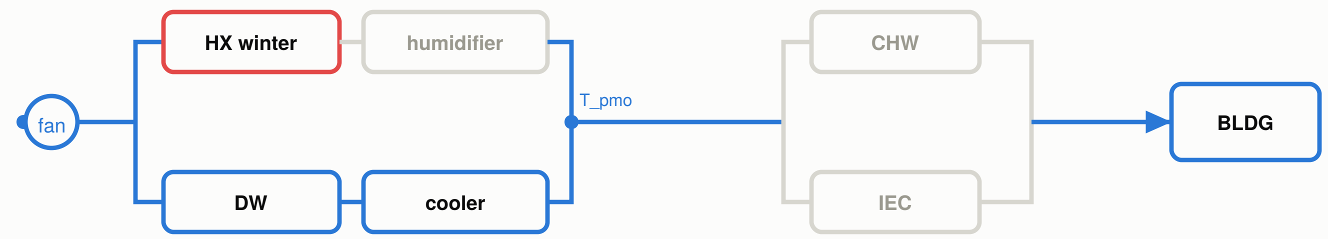


ABS + WCT · chilled water

dry CHW coil above dew point, then humidify — ORDER PROBLEM, see note

M6 heating + dehumidification

tier: — · geothermal water: ≥ 66 °C

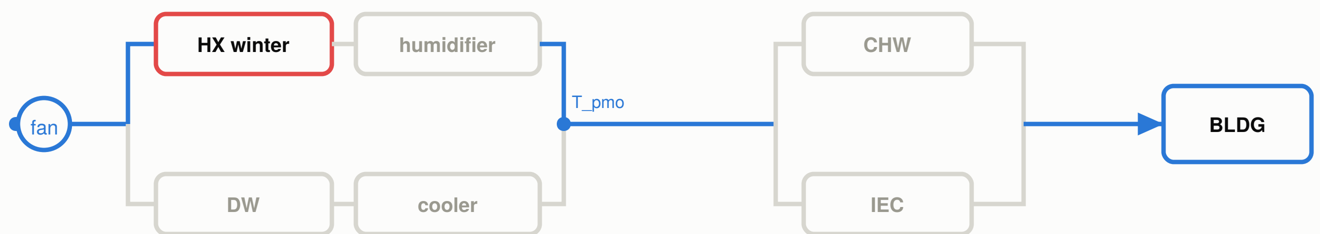


HX / hot water · scavenging loop

needs MODULATING d_pD / d_pE: dry one part, heat the other, blend

M7 heating + humidification

tier: mild · geothermal water: ≥ supply T

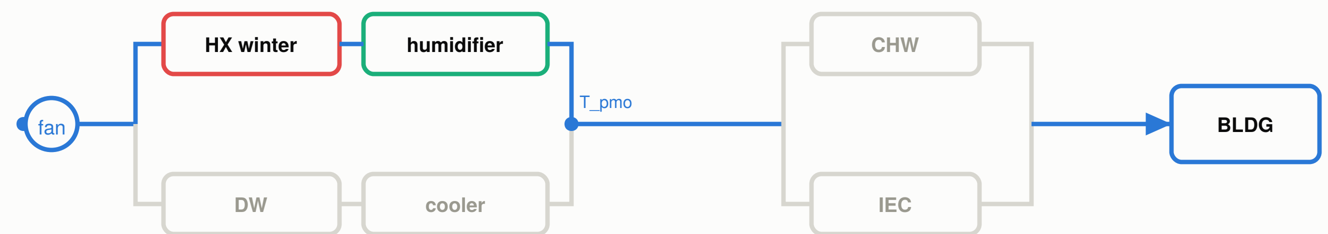


HX / hot water

heating only; latent deficit small enough to ignore

M8 heating + humidification

tier: full · geothermal water: ≥ supply T



HX / hot water

the winter workhorse — ~430 kW latent in January

Greyed components are present in the master architecture but inactive in that mode. Fork-1 dampers are binary except in M6.